



SEMESTER 1 EXAMINATIONS 2024/2025

MODULE: CSC1046 - OO Analysis & Design

PROGRAMME(S):

CASE	BSc Computer Apps (Soft.Eng.)
COMSCI	BSc Computer Science
ECSA	Study Abroad Program EC
ECSAO	Study Abroad Program EC

YEAR OF STUDY: 1,3

EXAMINER(S):

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TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer all questions.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

There are no additional requirements for this paper.

QUESTION 1

[TOTAL MARKS: 20]

Q 1(a)

[15 Marks]

Read the Appendix below, describing an Air ventilation Unit and complete the following:

Write three Use Cases to describe: *Setup System*; *Check CO2 level* ; *Check Heat*. Use Cockburn's approach in each case. Add extra variations and extensions as appropriate.

Q 1(b)

[5 Marks]

Draw a Use Case Diagram to represent your answer to 1(a) and add in two new use cases - one that is a <<use>> and one that <<extends>>.

[End of Question 1]

QUESTION 2

[TOTAL MARKS: 20]

Q 2(a)

[10 Marks]

Draw a Class Diagram to represent the Air Ventilation Unit in the appendix. Show two sample attributes and operations/functions for each class. Label all associations.

Q 2(b)

[5 Marks]

Explain and illustrate the difference between a dependency association and inheritance in your UML Class Diagram.

Q 2(c)

[5 Marks]

Show how to add a new *Light Sensor* to your Class Diagram.

[End of Question 2]

QUESTION 3**[TOTAL MARKS: 20]****Q 3(a)****[10 Marks]**

Draw a Sequence Diagram using the above Classes to model one of your Use Cases. Utilise the main features of UML Sequence Diagrams including highlighting when an object is computing, depiction of time constraints, return of values, creation and deletion of objects, representation of conditional behaviour and of iteration.

Q 3(b)**[10 Marks]**

Explain two of the SOLID principles with simple code examples.

[End of Question 3]**QUESTION 4****[TOTAL MARKS: 20]****Q 4(a)****[10 Marks]**

Use the Object Constraint Language to write an invariant for one of your classes class, in the style:

context <classifier>

inv [<constraint name>]: <Boolean OCL expression>

Q 4(b)**[5 Marks]**

Write a precondition for each of two functions that you have identified in any of your classes.

context <classifier>::<operation> (<parameters>)

pre [<constraint name>]: <Boolean OCL expression>

Q 4(c)**[5 Marks]**

Write a postcondition for each of two functions that you have identified in any of your classes.

context <classifier>::<operation> (<parameters>)

post [<constraint name>]: <Boolean OCL expression>

[End of Question 4]

QUESTION 5**[TOTAL MARKS: 20]****Q 5(a)****[14 Marks]**

Draw two separate State Diagrams to represent two of your objects in the Air Ventilation Unit. Ensure that you represent all of the transitions correctly with guards.

Q 3(b)**[6 Marks]**

Show how parallelism can be included in your diagrams and why the creation of sub- and super-states can be useful.

[End of Question 5]**APPENDIX****Air Ventilation Unit**

The brief specification of a new air ventilation unit would include three main subsystems:

1. A set of sensors for Heat, CO₂ (Carbon Dioxide) and Ozone.
2. An air input fan system that can be set to HIGH, LOW and OFF.
3. A Heating Element that can be set to 20°, 16°, 14°, Celsius or OFF

The system will have a control panel allowing the setting of initial values but once running the system should work automatically to control the temperature and air quality based on the readings from the sensors.

Standard cut-off points would be that CO₂ should stay between 400 and 800 parts per million. If it goes beyond 800 the fan should move to HIGH until it drops to 600. It should then drop to LOW, it should stay in LOW until it reaches 400 then it should turn to OFF.

If the temperature drops 2 degrees below its setting the heat should turn on until it is 2 degrees above its setting.

Ozone levels should be below 0.08 parts per million (ppm) again the fan should be switched on to maintain this.

Finally, if after the Fan is on HIGH for 20 minutes and the ozone or CO₂ have not dropped below their safe levels an alarm should sound to indicate that the room should be vacated.

[END OF EXAM]